

Entity Relationship-Model

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Introduction of ER Model

- ▶ Peter Chen developed the ER diagram in 1976 .
- ▶ The ER model was created to provide a simple and understandable model for representing the structure and logic of databases.
- ▶ It has since evolved into variations such as the Enhanced ER Model and the Object Relationship Model.
- ▶ The Entity Relational Model is a model for identifying entities to be represented in the database and representation of how those entities are related.
- ▶ The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.

Entity-Relationship Model

- ▶ An Entity-Relationship Model represents the structure of the database with the help of a diagram.
- ▶ ER Modelling is a systematic process to design a database as it would require you to analyze all data requirements before implementing your database.

Why Use ER Diagrams In DBMS?

- ▶ ER diagrams are used to represent the E-R model in a database, which makes them easy to convert into relations (tables).
- ▶ ER diagrams provide the purpose of real-world modeling of objects which makes them intently useful.
- ▶ ER diagrams require no technical knowledge and no hardware support.
- ▶ These diagrams are very easy to understand and easy to create even for a naive user.
- ▶ It gives a standard solution for visualizing the data logically.
- ▶ It reduces complexity and allows database designers to build databases quickly

Entity Relationship Diagram

- ▶ ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:
- ▶ **An Entity Relationship Diagram is a diagram that represents relationships among entities in a database.**
- ▶ **It is commonly known as an ER Diagram. An ER Diagram in DBMS plays a crucial role in designing the database.**
- ▶ **Today's business world previews all the requirements demanded by the users in the form of an ER Diagram.**
- ▶ **Later, it's forwarded to the database administrators to design the database.**

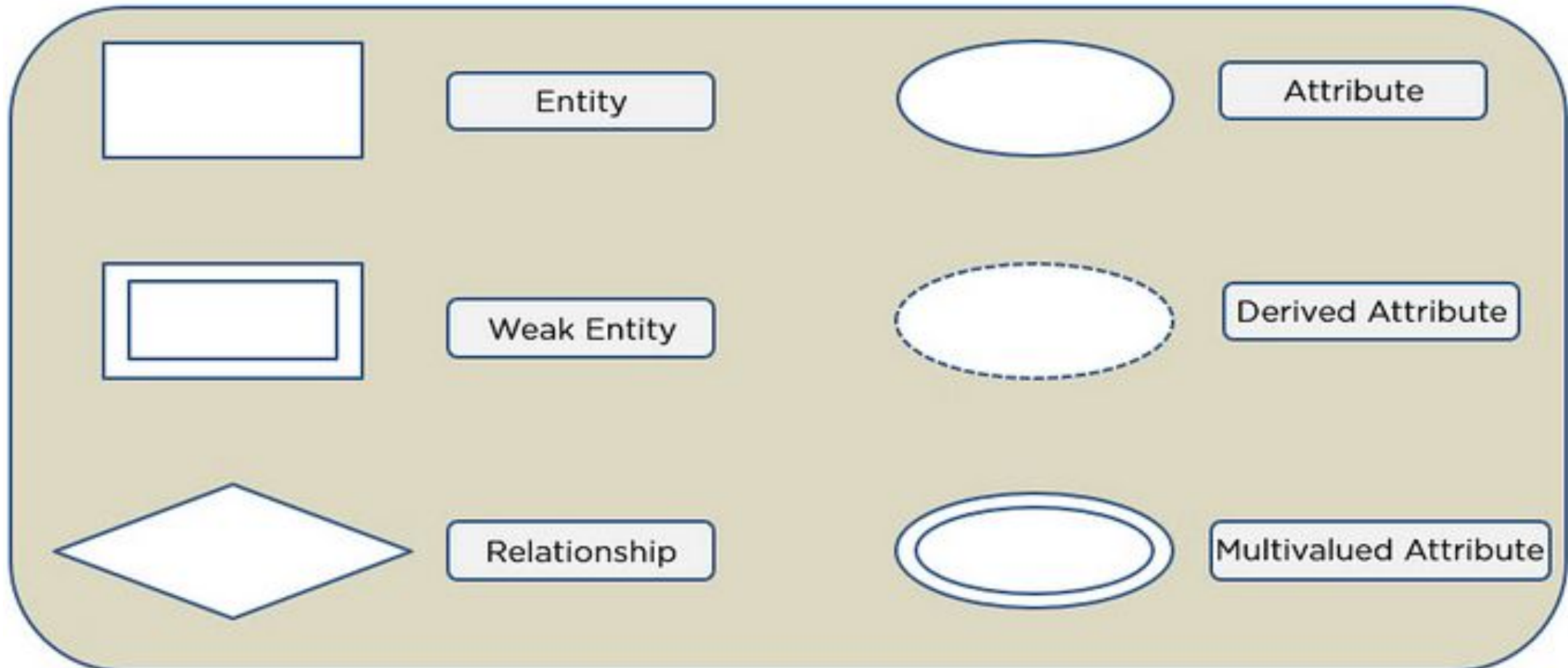
Symbols Used in ER Model

- ▶ ER Model is used to model the logical view of the system from a data perspective which consists of these symbols:
- ▶ **Rectangles:** Rectangles represent Entities in the ER Model.
- ▶ **Ellipses:** Ellipses represent Attributes in the ER Model.
- ▶ **Diamond:** Diamonds represent Relationships among Entities.
- ▶ **Lines:** Lines represent attributes to entities and entity sets with other relationship types.
- ▶ **Double Ellipse:** Double Ellipses represent Multi-Valued Attributes.
- ▶ **Double Rectangle:** Double Rectangle represents a Weak Entity.

What is an ER Diagram?

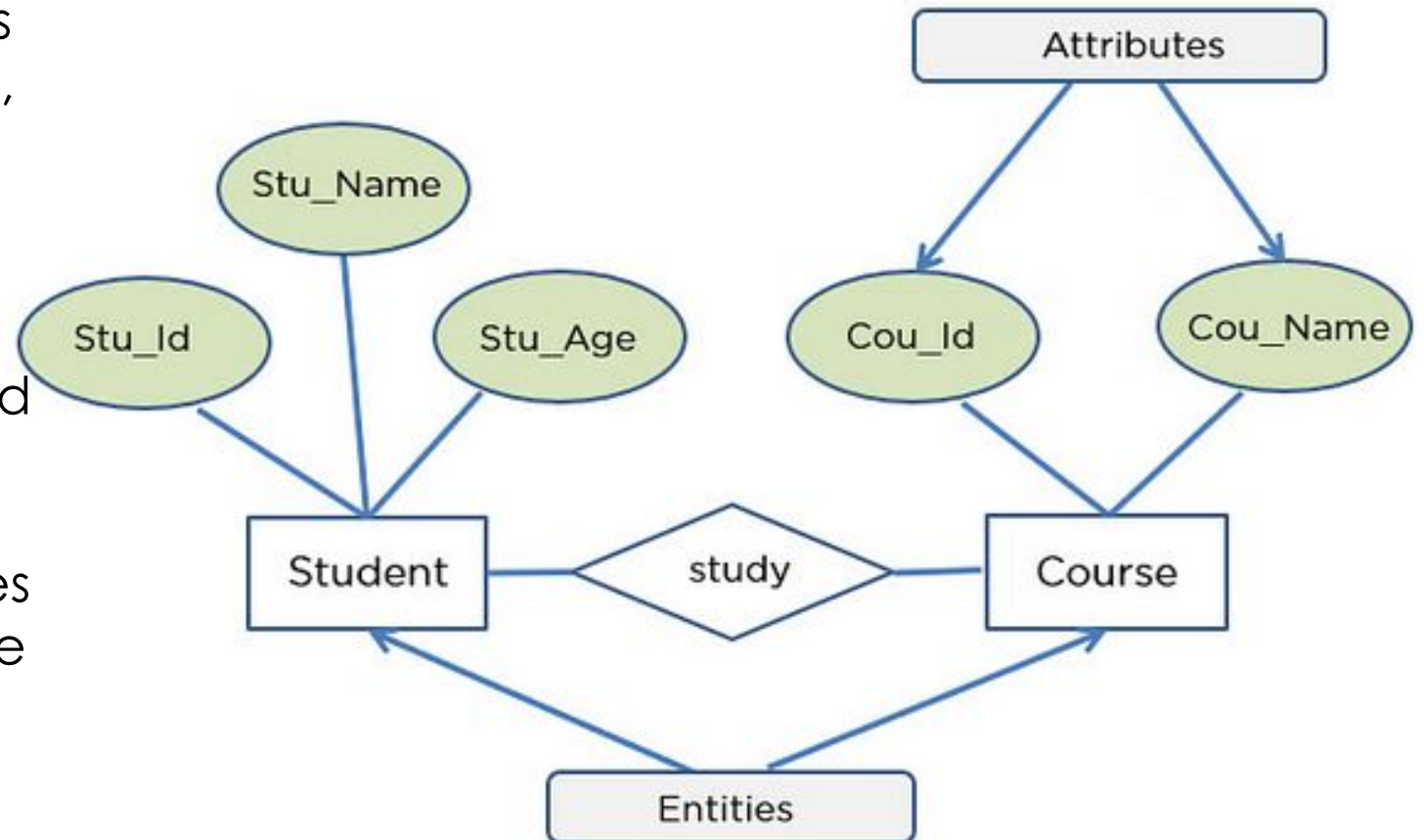
- ▶ An Entity Relationship Diagram (ER Diagram) pictorially explains the relationship between entities to be stored in a database.
- ▶ Fundamentally, the ER Diagram is a structural design of the database. It acts as a framework created with specialized symbols for the purpose of defining the relationship between the database entities.
- ▶ ER diagram is created based on three principal components: entities, attributes, and relationships.

Symbols Used in ER Diagrams



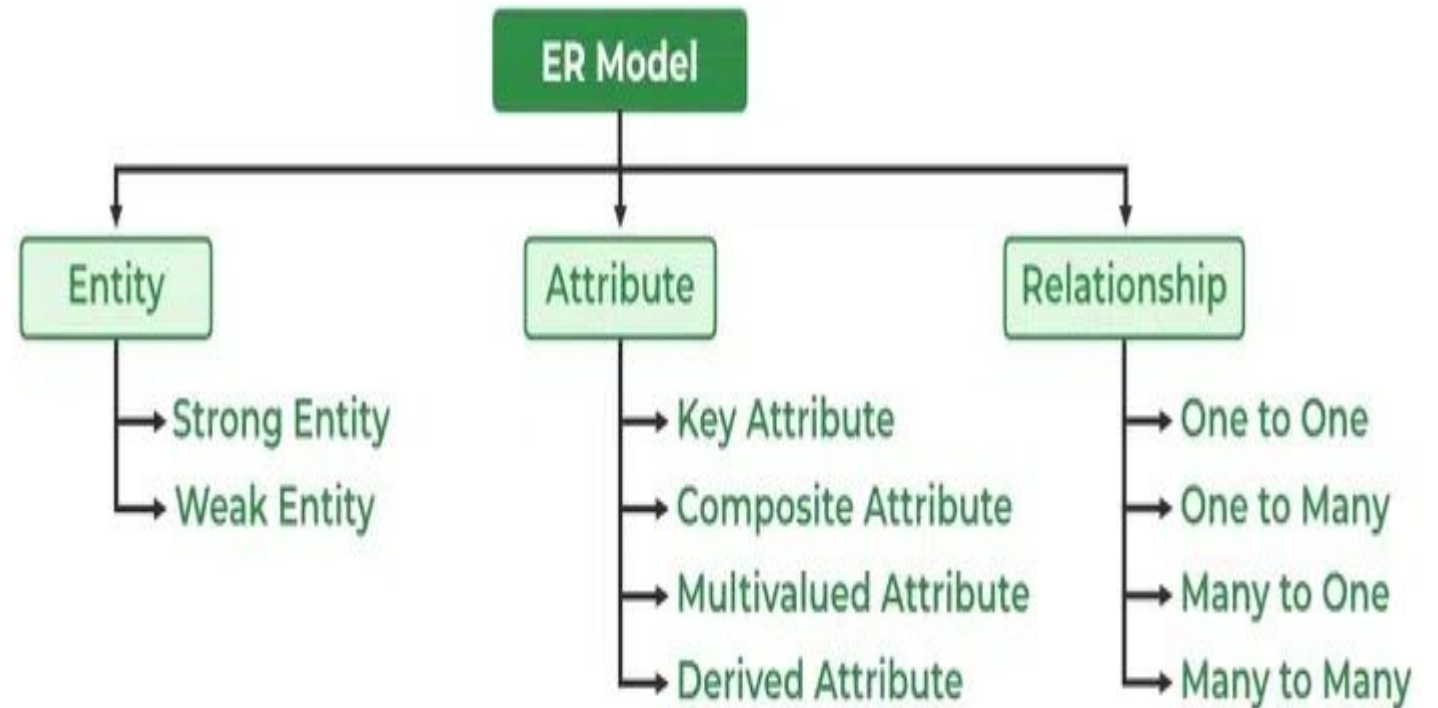
Example: E-R diagram

- The following diagram showcases two entities - Student and Course, and their relationship.
- The relationship described between student and course is many-to-many, as a course can be opted by several students, and a student can opt for more than one course.
- Student entity possesses attributes - Stu_Id, Stu_Name & Stu_Age. The course entity has attributes such as Cou_ID & Cou_Name.



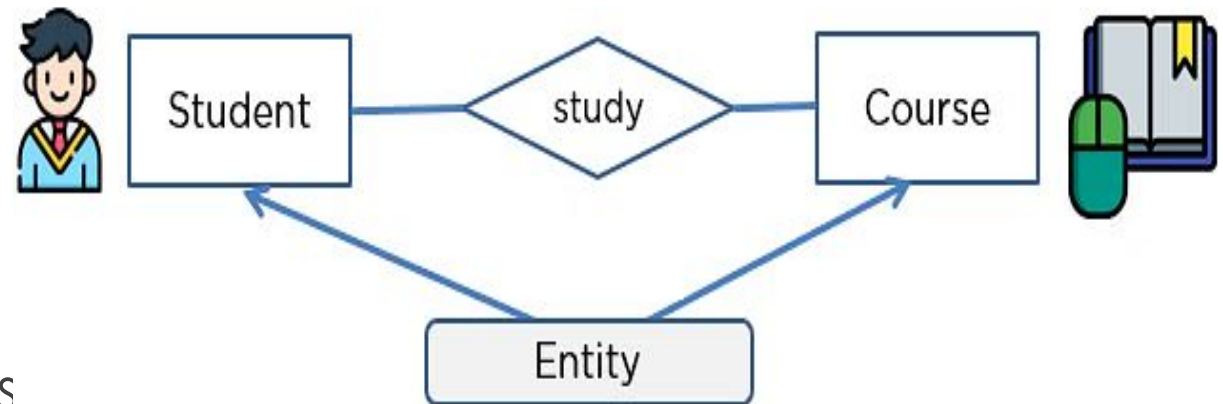
Components of ER Diagram

- ▶ You base an ER Diagram on three
- ▶ Entities
 - ▶ Weak Entity
- ▶ Attributes
 - ▶ Key Attribute
 - ▶ Composite Attribute
 - ▶ Multivalued Attribute
 - ▶ Derived Attribute
- ▶ Relationships
 - ▶ One-to-One Relationships
 - ▶ One-to-Many Relationships
 - ▶ Many-to-One Relationships
 - ▶ Many-to-Many Relationships



Entities

- ▶ An entity can be either a living or non-living component.
- ▶ It showcases an entity as a rectangle in an ER diagram.
- ▶ For example, in a student study course, both the student and the course are entities.
- ▶ An Entity may be an object with a physical existence – a particular person,
 - ▶ car, house, or
 - ▶ employee – or it
 - ▶ may be an object with
 - ▶ a conceptual existence –
 - ▶ a company, a job, or a university cours



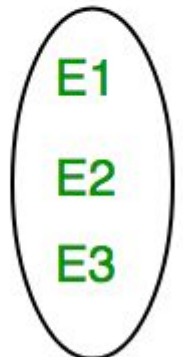
Entity Set:

- ▶ Entity Set: An Entity is an object of Entity Type and a set of all entities is called an entity set.
- ▶ For Example, E1 is an entity having Entity Type Student and the set of all students is called Entity Set. In ER diagram, Entity Type is represented as:
- ▶ 1. Strong Entity
- ▶ A Strong Entity is a type of entity that has a key Attribute. Strong Entity does not depend on other Entity in the Schema. It has a primary key, that helps in identifying it uniquely, and it is represented by a rectangle. These are called Strong Entity Types.
- ▶ 2. Weak Entity
- ▶ An Entity type has a key attribute that uniquely identifies each entity in the entity set. But some entity type exists for which key attributes can't be defined. These are called Weak Entity types.



Student

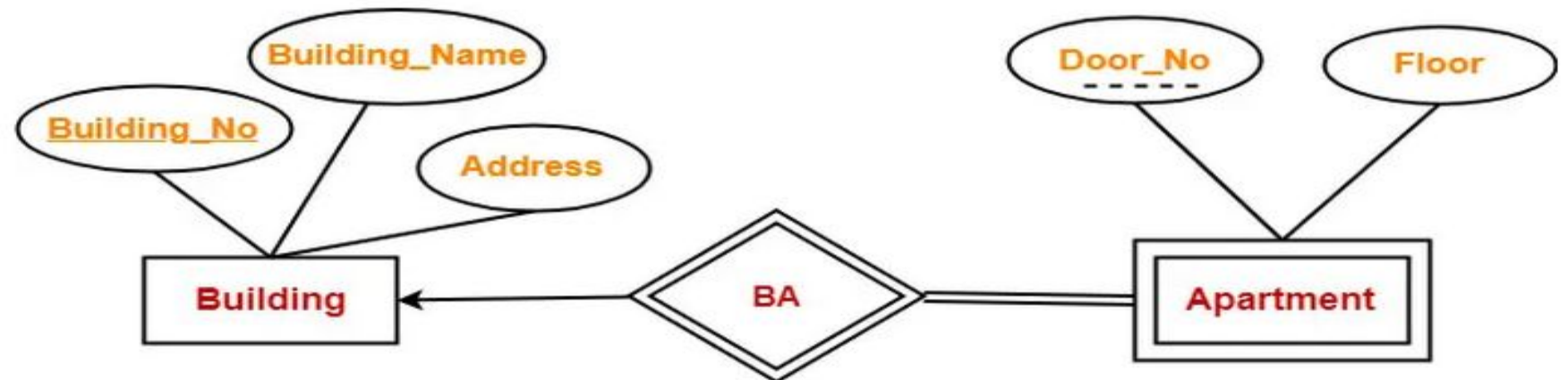
Entity Type



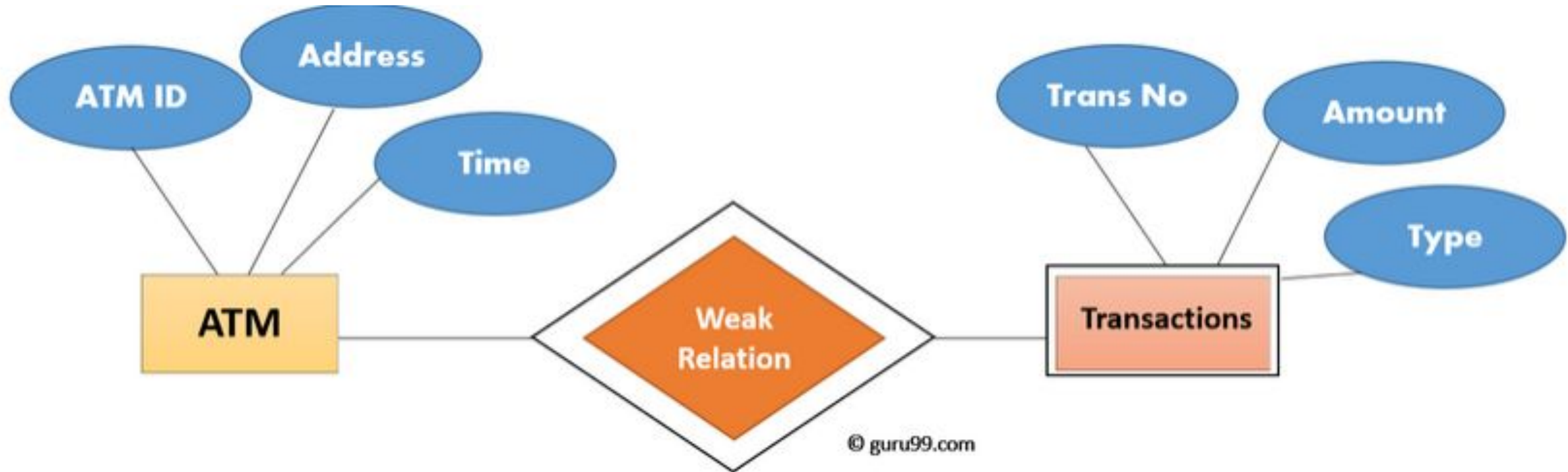
Entity Set

A weak entity

- ▶ A weak entity type is represented by a Double Rectangle. The participation of weak entity types is always total.
- ▶ The relationship between the weak entity type and its identifying strong entity type is called identifying relationship and it is represented by a double diamond.
- ▶ A weak entity depends on another entity to exist. It does not have a primary key but a partial key, which helps identify its instances in combination with the identifying entity's primary key.



Example of Weak Entity



Attributes

- ▶ Attributes are the properties that define the entity type. For example, Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student. In ER diagram, the attribute is represented by an oval.



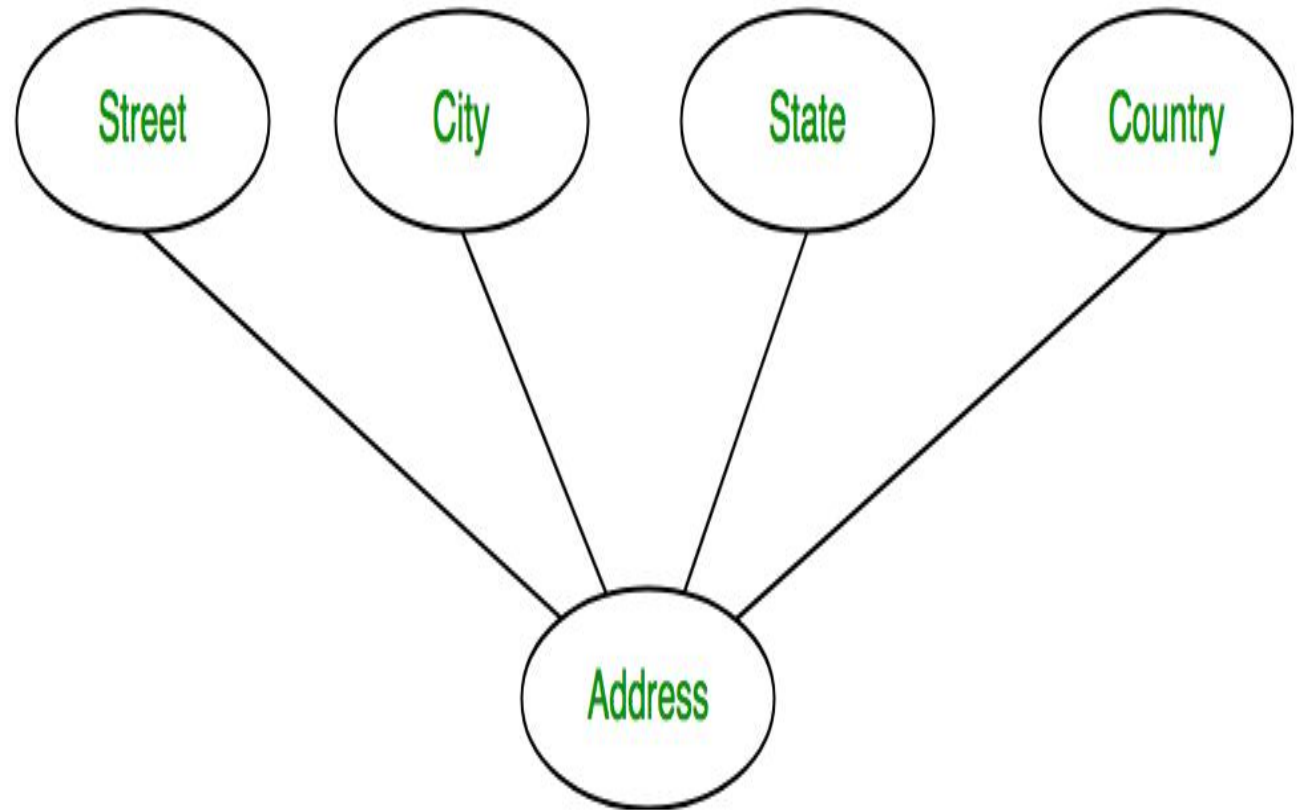
Key Attribute

- ▶ The attribute which uniquely identifies each entity in the entity set is called the key attribute.
- ▶ For example, Roll_No will be unique for each student. In ER diagram, the key attribute is represented by an oval with underlying lines.



Composite Attribute

- ▶ An attribute composed of many other attributes is called a composite attribute.
- ▶ For example, the Address attribute of the student Entity type consists of Street, City, State, and Country.
- ▶ In ER diagram, the composite attribute is represented by an oval comprising of ovals.



Multivalued Attribute

- ▶ An attribute consisting of more than one value for a given entity.
- ▶ For example, Phone_No (can be more than one for a given student).
- ▶ In ER diagram, a multivalued attribute is represented by a double oval.

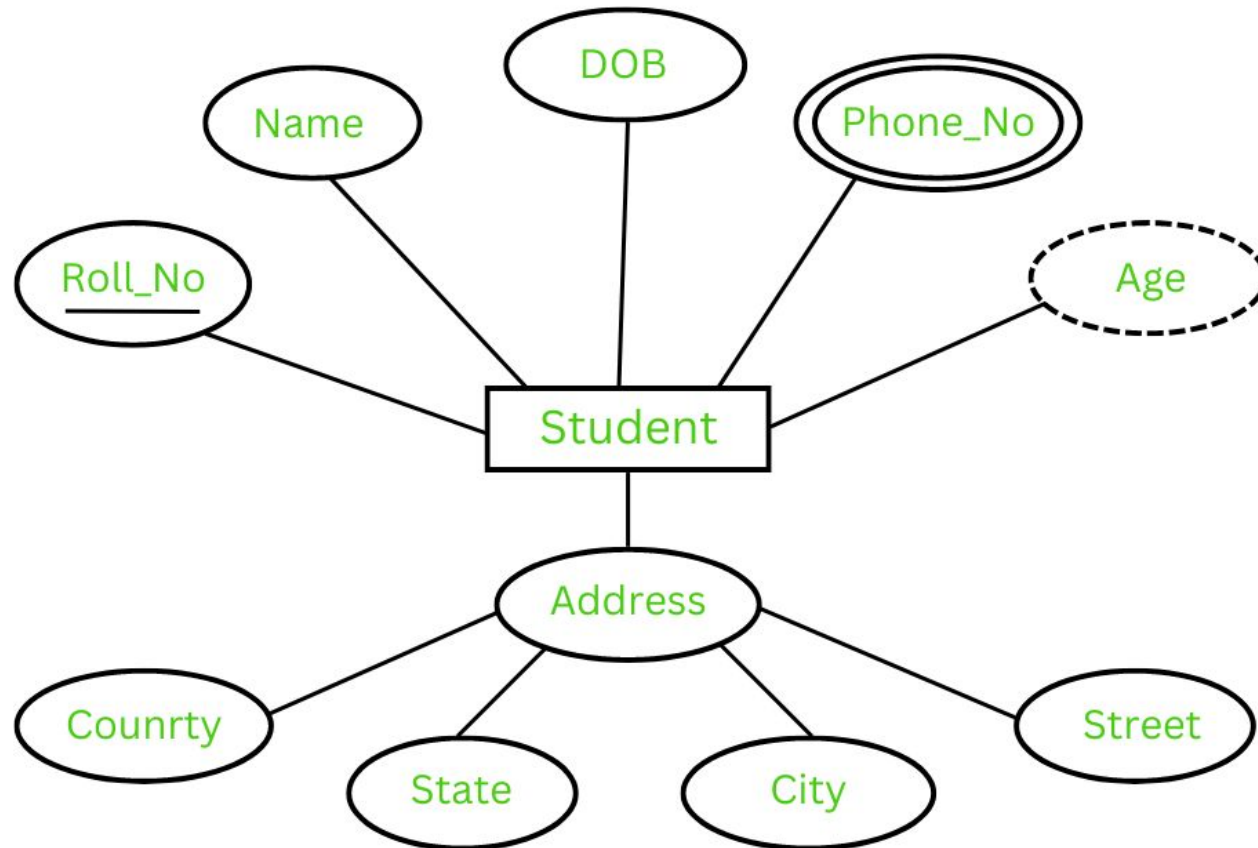


Derived Attribute

- ▶ An attribute that can be derived from other attributes of the entity type is known as a derived attribute. e.g.; Age (can be derived from DOB).
- ▶ In ER diagram, the derived attribute is represented by a dashed oval.



The Complete Entity Type Student with its Attributes can be represented as:



Relationship

- ▶ A **Relationship** is a logical link between two or more entities, depicting how the entities interact with each other. Relationships help to define the associations and dependencies between entities in a database.
- ▶ **There are four major types of relationships:**
- ▶ **1. One to One Relationship**
 - ▶ In this type of relationship, a single instance of an entity A is associated with a single instance of entity B. For instance, in a company database, an entity 'Employee' can have a one-to-one relationship with an entity 'EmployeeDetails', where each employee has one set of details.
- ▶ **2. One to Many:** This is a relationship where a single instance of an entity A can be associated with multiple instances of entity B, but each instance of entity B can be associated with only one instance of entity A. For example, a 'Department' entity can have a one-to-many relationship with an 'Employee' entity, representing that one department can have many employees, but each employee belongs to one department.
- ▶ **3. Many to One:** This is the inverse of the one-to-many relationship, where many instances of entity A can be associated with a single instance of entity B.
- ▶ **4. Many to Many:** In this relationship, many instances of entity A can be associated with many instances of entity B. For example, an 'Author' entity can have a many-to-many relationship with a 'Book' entity, indicating that one author can write many books, and one book can have many authors.

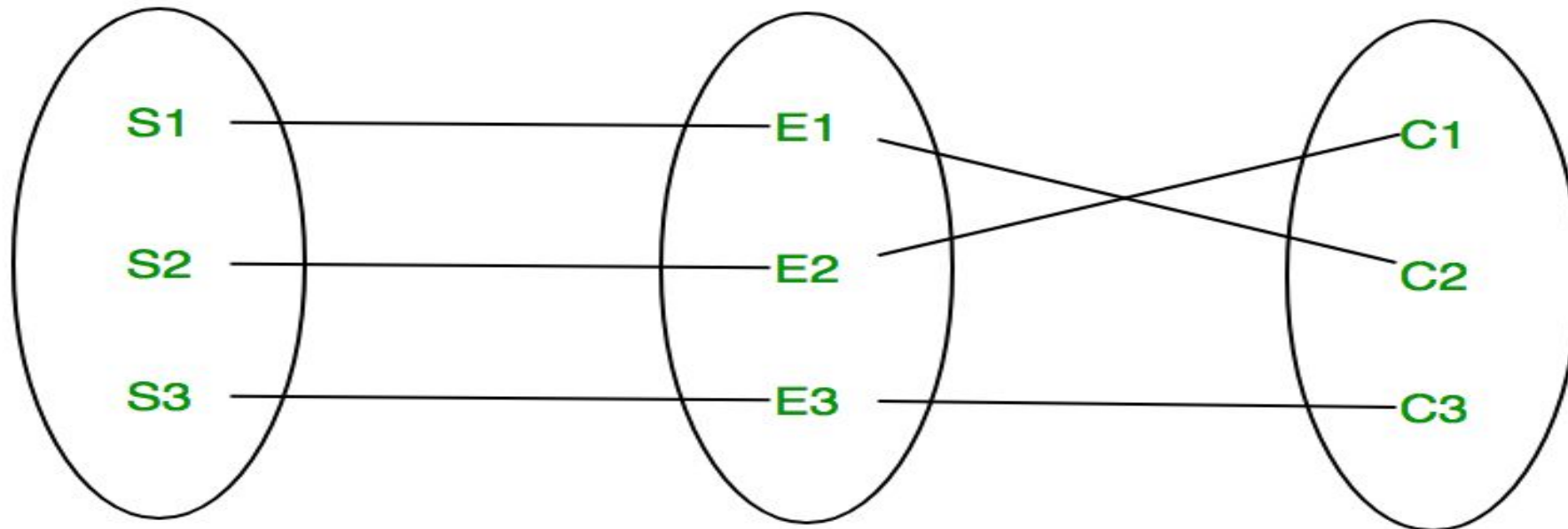
Relationship Type and Relationship Set

- ▶ A Relationship Type represents the association between entity types.
- ▶ For example, 'Enrolled in' is a relationship type that exists between entity type Student and Course.
- ▶ In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.



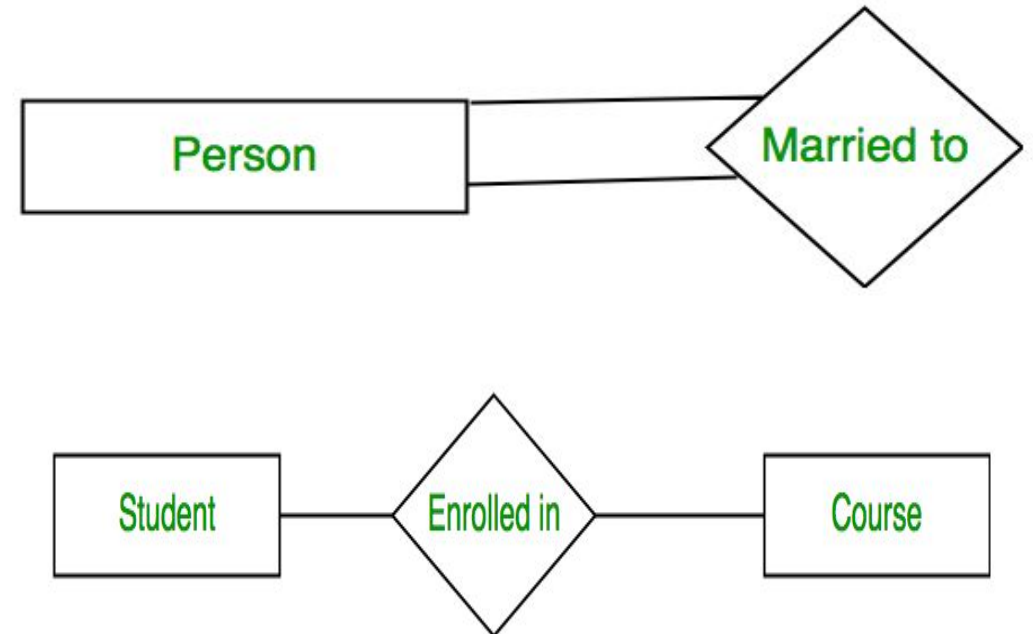
relationship set

- ▶ A set of relationships of the same type is known as a relationship set.
- ▶ The following relationship set depicts S1 as enrolled in C2, S2 as enrolled in C1, and S3 as registered in C3.



Degree of a Relationship Set

- ▶ The number of different entity sets participating in a relationship set is called the degree of a relationship set.
- ▶ **1. Unary Relationship:** When there is only ONE entity set participating in a relation, the relationship is called a unary relationship. For example, one person is married to only one person.
- ▶ **Binary Relationship:** When there are TWO entities set participating in a relationship, the relationship is called a binary relationship. For example, a Student is enrolled in a Course.
- ▶ **Ternary Relationship:** When there are n entities set participating in a relation, the relationship is called an n-ary relationship.



Binary Relationship:

- ▶ Binary Relationship: A binary relationship exists when exactly two entity type participates. When such a relationship is present we say that the degree is 2.

- ▶ This is the most common degree of relationship. It is easy to deal with such relationship as these can be easily converted into relational tables.

For example, we have two entity type 'Customer' and 'Account' where each

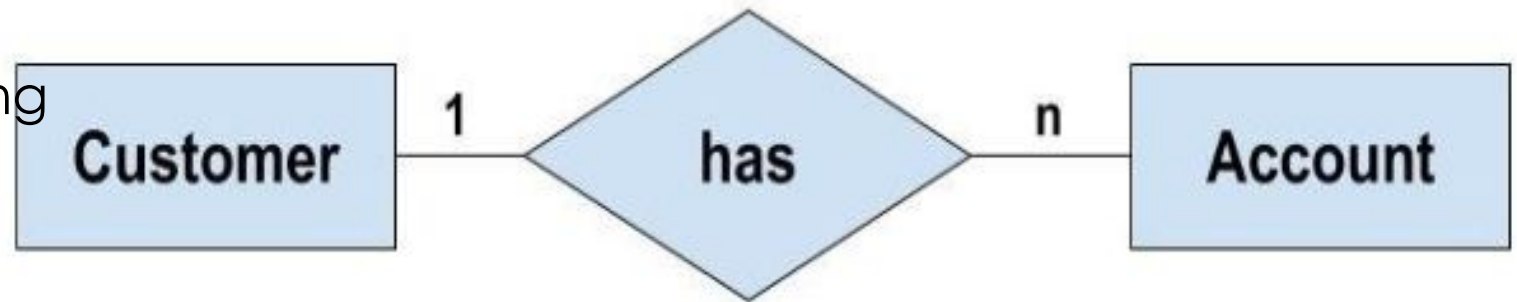
'Customer' has an 'Account' which stores the account details of the 'Customer'. Since

we have two entity types participating we call it a binary relationship.

Also, one

'Customer' can have many 'Account' but each 'Account' should belong to only one 'Customer'.

We can say that it is a one-to-many binary relationship.

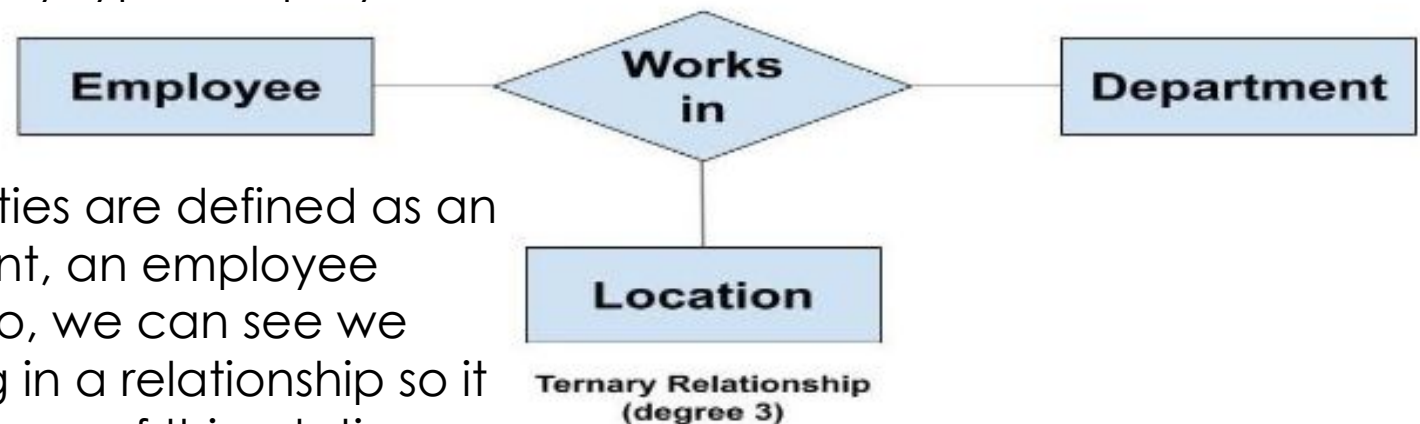


Binary Relationship
(degree 2)

Ternary Relationship:

- ▶ A ternary relationship exists when exactly three entity type participates. When such a relationship is present we say that the degree is 3.
- ▶ As the number of entity increases in the relationship, it becomes complex to convert them into relational tables.

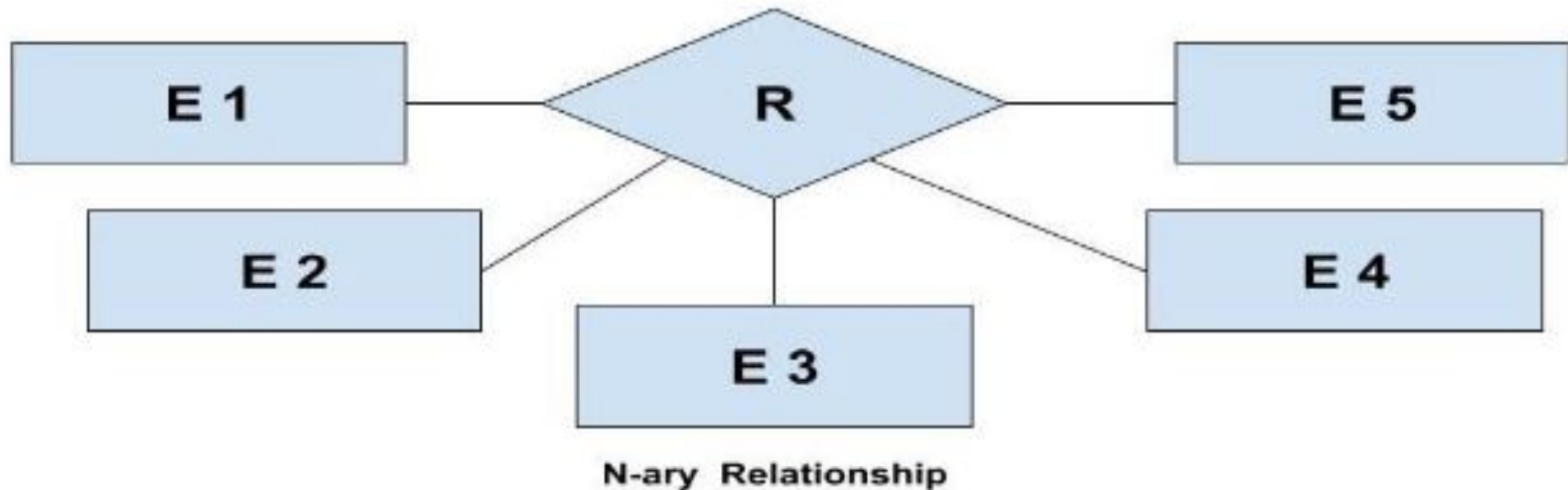
For example, we have three entity type 'Employee', 'Department' and 'Location'.



relationship between these entities are defined as an employee works in a department, an employee works at a particular location. So, we can see we have three entities participating in a relationship so it is a ternary relationship. The degree of this relation is 3

n-ary Relationship:

- ▶ An N-ary relationship exists when 'n' number of entities are participating. So, any number of entities can participate in a relationship.
- ▶ There is no limitation to the maximum number of entities that can participate.



Mapping Constrains

- ▶ For binary relationship set R on an entity set A and B , there are four possible mapping cardinalities.
- ▶ a. One-to-One (1:1)
- ▶ b. One-to-Many (1:M)
- ▶ c. Many-to-One (M:1)
- ▶ d. Many-to-Many (M:M)

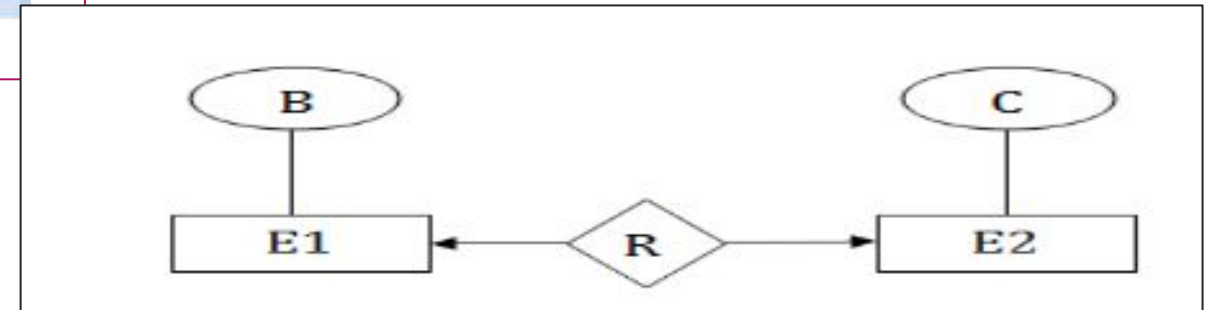
One-to-One Relationship



- ▶ When a single element of an entity is associated with a single element of another entity, it is called a one-to-one relationship.
- ▶ When only one instance of an entity is associated with the relationship, then it is known as one to one relationship.
- ▶ For example, a student has only one identification card and an identification card is given to one person.



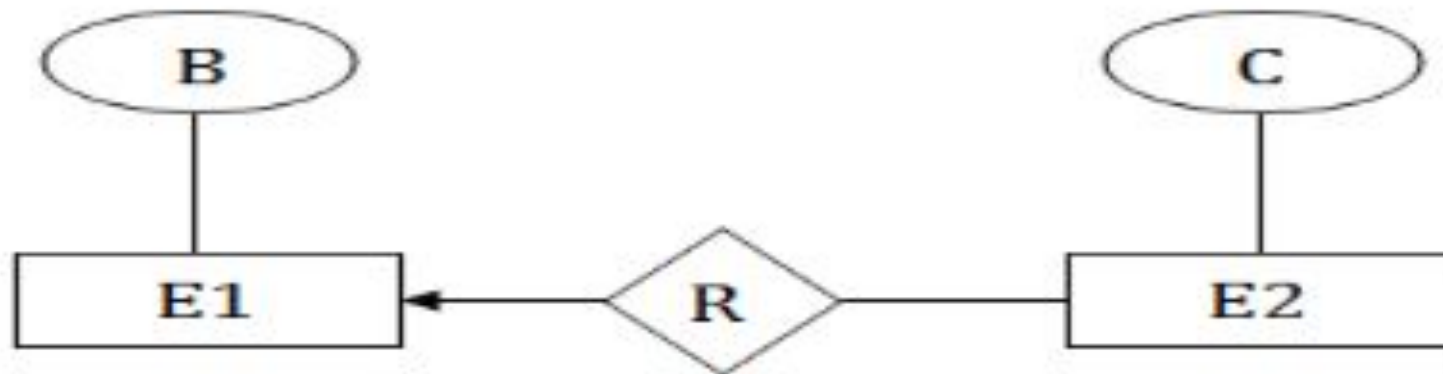
or example: A female can marry to one male, and a male can marry to one female.



One-to-many relationship:



- ▶ When only one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then this is known as a one-to-many relationship.
- ▶ For example, Scientist can invent many inventions, but the invention is done by the only specific scientist.



One-Many to Relationship

Customer_id	Customer_name	Customer_n0
111	Maria Danzie	1199229921
222	Alex Brat	3939637382
333	Sania Martini	8249283563

Order_id	Order_amount	Customer_id
10001	1200	222
10002	2000	333
10003	4500	222
10004	1220	111
10005	3550	222



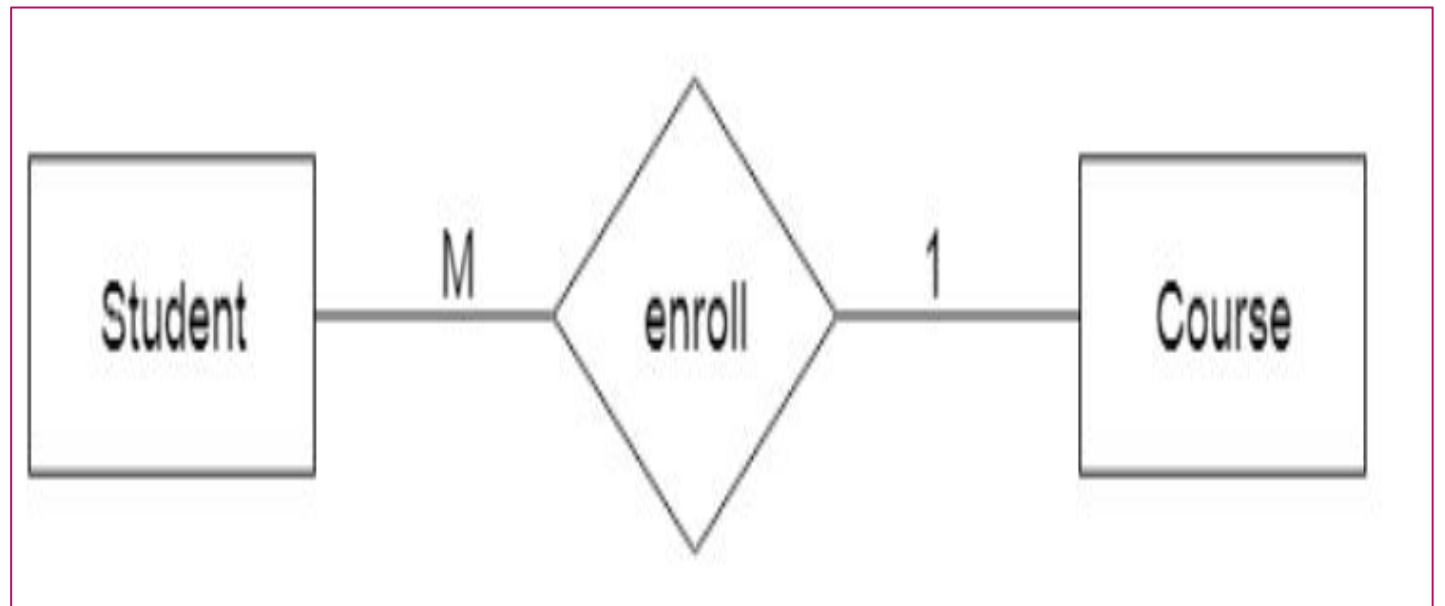
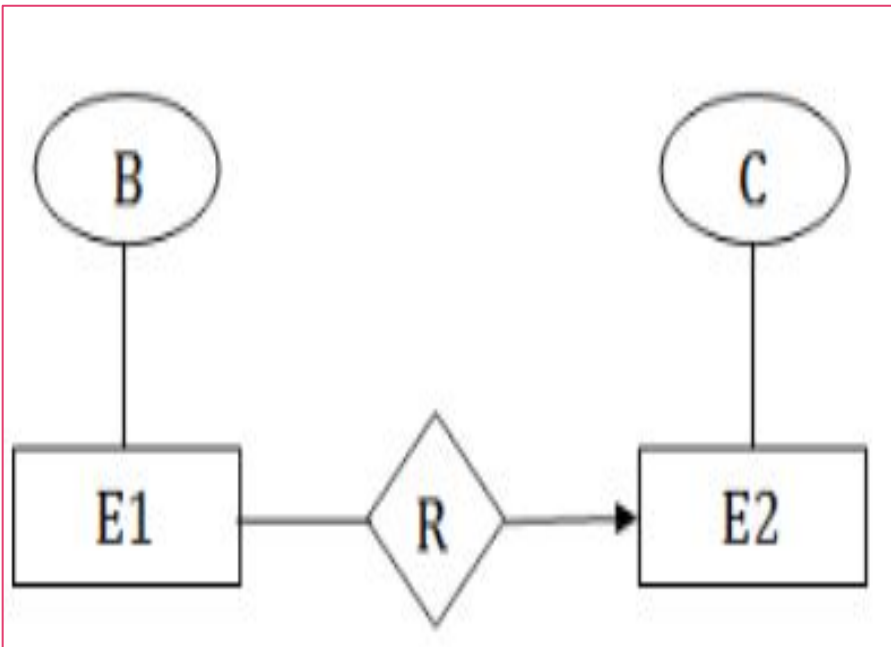
Table A

Table B



Many-to-one relationship:

- ▶ When more than one instance of the entity on the left, and only one instance of an entity on the right associates with the relationship then it is known as a many-to-one relationship.
- ▶ For example, Student enrolls for only one course, but a course can have many students.



Many-to-many relationship:

- ▶ When more than one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then it is known as a many-to-many relationship.
- ▶ For example, Employee can assign by many projects and project can have many employees



Many-to-Many Relationship

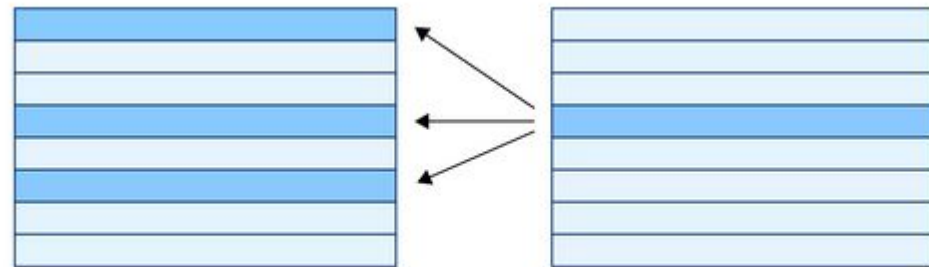


Table A

Table B

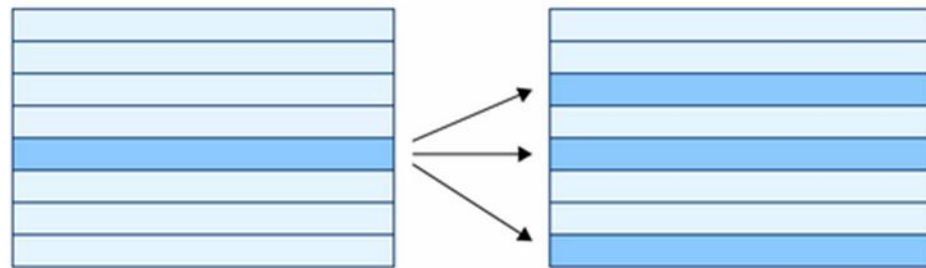


Table A

Table B

Order_id	Order_amount	Customer_id
10001	1200	222
10002	2000	333
10003	4500	222
10004	1220	111
10005	3550	222

Order_id	Item_id
10001	1201
10001	1203
10004	1202
10004	1203
10005	1201
10003	1201

Item_id	Item_name
1201	Maggie
1202	Pizza
1203	Kurtossh

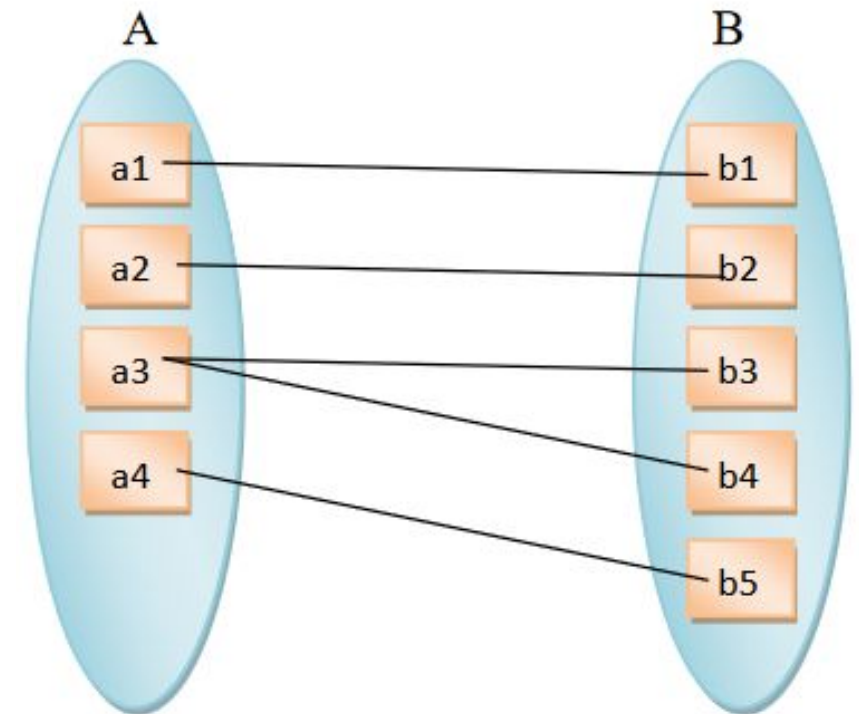
Participation Constraints

- ▶ The participation constraint specifies the number of instances of an entity can participate in a relationship set
- ▶ Types of Participation Constraints

1. Total Participation
2. Partial Participation

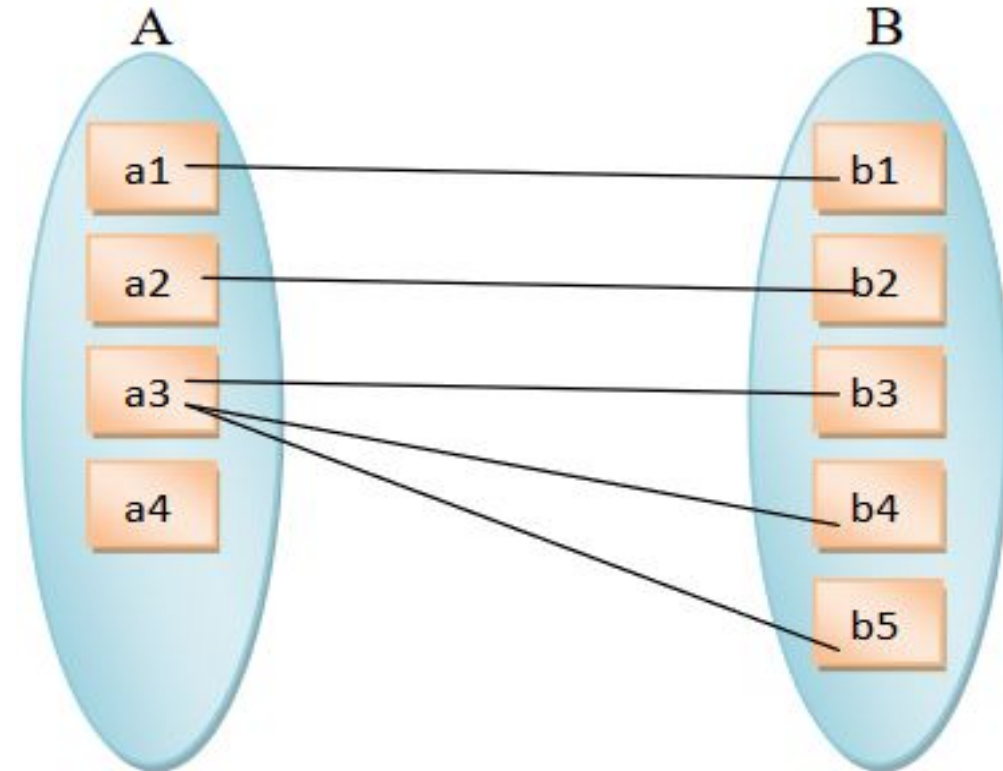
Total Participation

- ▶ **The Participation of an entity set E in a relationship set R is said to be total if every entity in E participates in at least one relationship in R.**
- ▶ The participation of entity set A in the relationship set is total because every entity of A participates in the relationship set
- ▶ and
- ▶ The participation of entity set B in the relationship set is also total because every entity of B also participates in the relationship Set



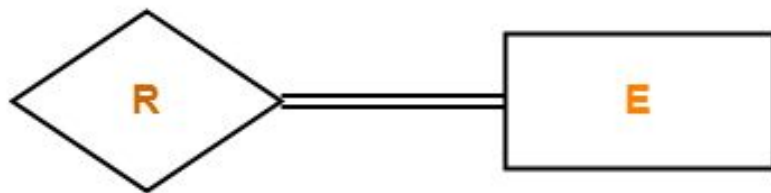
Partial Participation

- ▶ The participation of an entity set E in relationship set R is said to be partial if only some entities in E participate in relationships in R .
- ▶ The participation of entity set A in the relationship set is partial because only some entities of A participate in the relationship set.
- ▶ while
- ▶ The participation of entity set B in the relationship set is total because every entity of B participates in the relationship set.

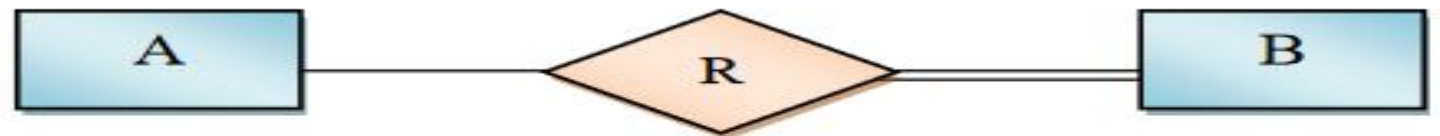


The total participation Example

- ▶ In ERD, the total participation is denoted by doubled-line between entity set and relationship set and partial participation is denoted by single line between entity set and relationship set.
- ▶ That is why, it is also called as **mandatory participation**.



Total Participation



Participation in R: **Partial A**
Total B

Example:

- ▶ Suppose an entity set Student related to an entity set Course through Enrolled relationship set.
- ▶ The participation of entity set course in enrolled relationship set is partial because a course may or may not have students enrolled in.
- ▶ It is possible that only some of the course entities are related to the student entity set through the enrolled relationship set.
- ▶ The participation of entity set student in enrolled relationship set is total because every student is expect to relate at least one course through the enrolled relationship set



Participation in Enrolled relationship set: **Partial** Course
Total Student

partial participation

- ▶ Here,
- ▶ Single line between the entity set “Course” and relationship set “Enrolled in” signifies partial participation.
- ▶ It specifies that there might exist some courses for which no enrollments are made.



Participation in Enrolled relationship set: **Partial** Course
Total Student

Cardinality Constraints

- ▶ Here cardinality represents the number of times an entity of an entity set participates in a relationship set. Or we can say that the cardinality of a relationship is the number of tuples (rows) in a relationship.
- ▶ A **one-to-one** :example would be one student associated with one mailing address.
- ▶ A **one-to-many** :example (or many-to-one, depending on the relationship direction): One student registers for multiple courses, but all those courses have a single line back to that one student.
- ▶ **Many-to-many** :example: Students as a group are associated with multiple faculty members, and faculty members in turn are associated with multiple students.

Relationship between Cardinality and Participation Constraints-

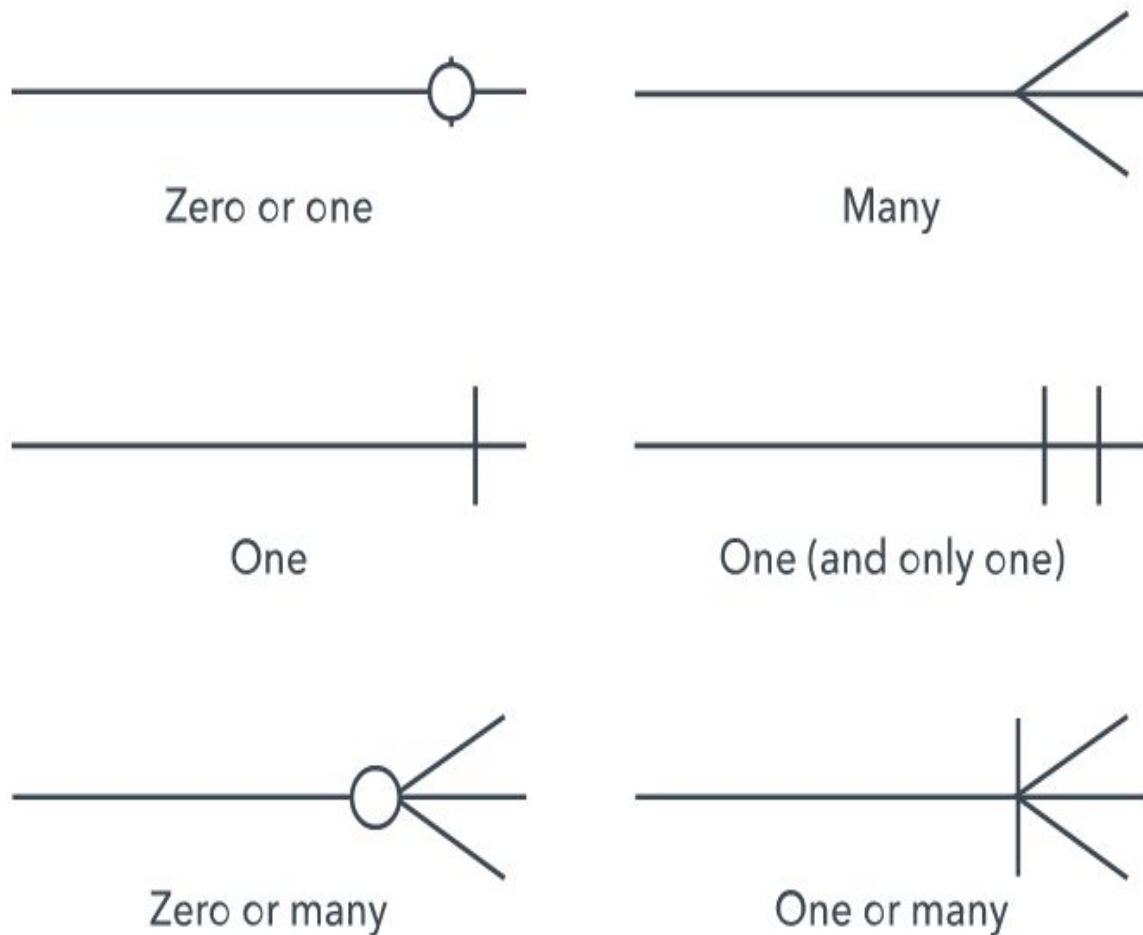
- ▶ **Structural Constraints** : Structural Constraints are also called Structural properties of a database management system (DBMS).
- ▶ Cardinality Ratios and Participation Constraints taken together are called Structural Constraints.
- ▶ Minimum cardinality tells whether the participation is partial or total.
- ▶ If minimum cardinality = 0, then it signifies partial participation.
- ▶ If minimum cardinality = 1, then it signifies total participation.
- ▶ Maximum cardinality tells the maximum number of entities that participates in a relationship set.

Let's see an example to understand this ER Model based on the information above.

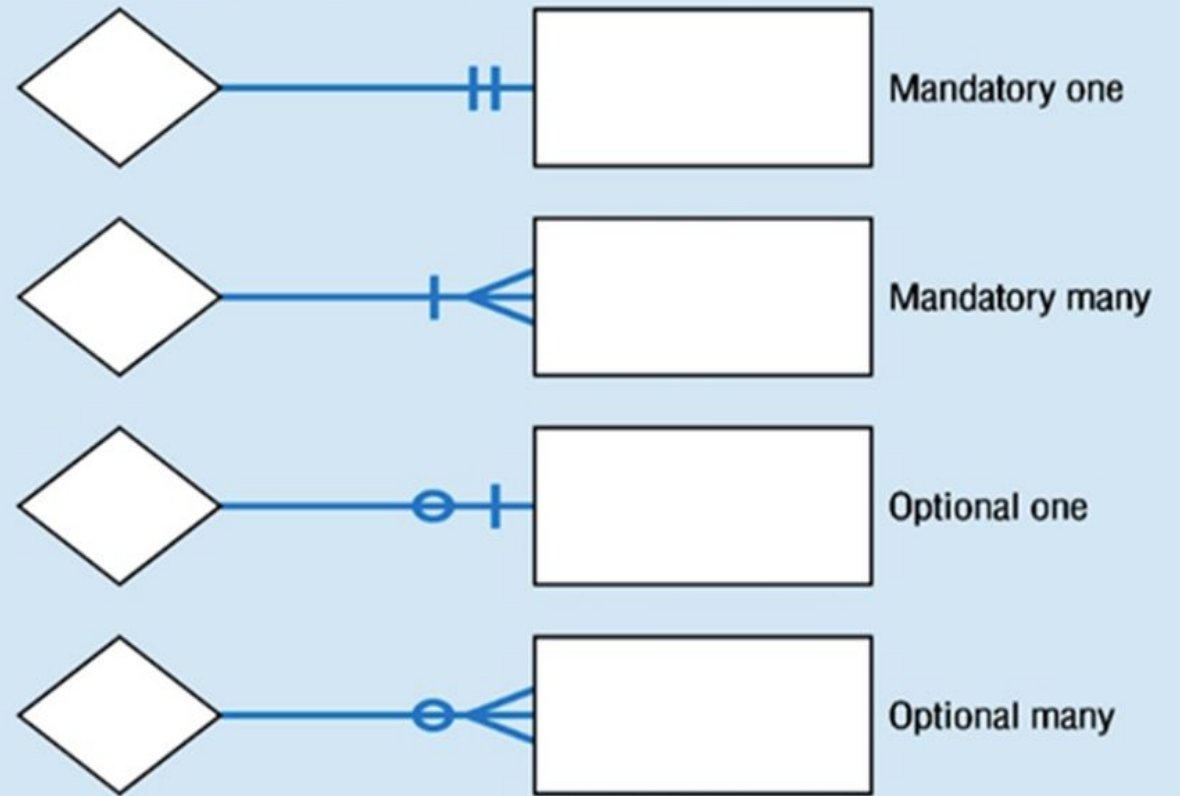
- ▶ Let's take an **e-commerce platform** example. The goal is to illustrate how different components (entities) in a typical e-commerce business can interact with each other and what kind of attributes they might have. This ER model helps to depict the logical structure and flow of data in the system.
- ▶ **Entities**
 1. **Customer**
 - ▶ This entity represents the customers who are using the e-commerce platform.
- ▶ **Attributes**

Cardinality:

Show whether the relationship is 1-1, 1-many or many-to-many.



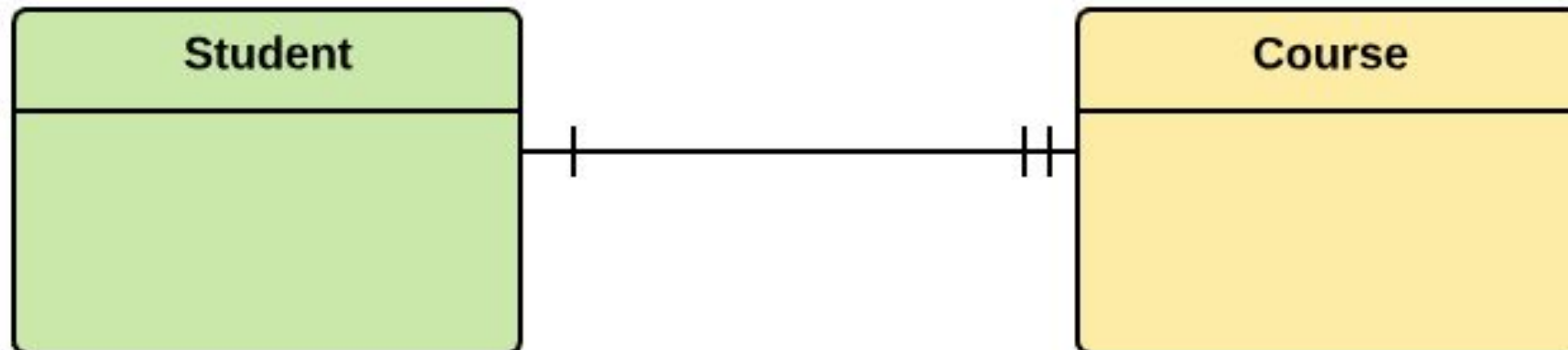
Relationship cardinality



1.One-to-one:

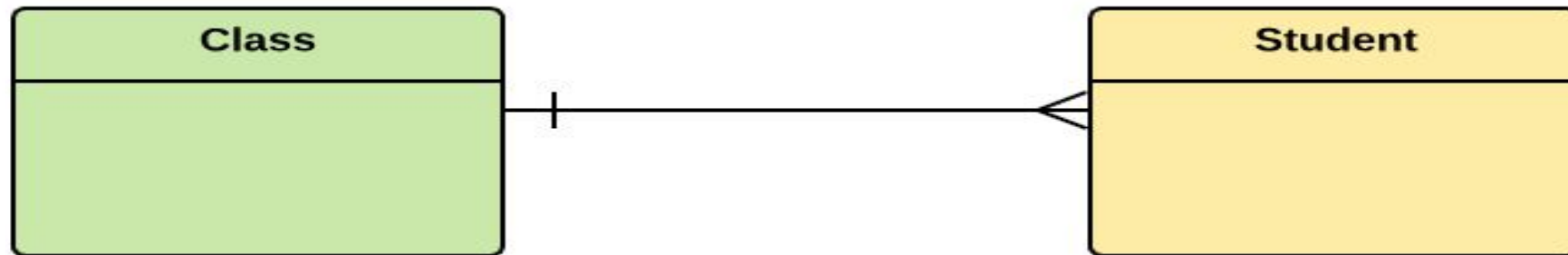
One entity from entity set X can be associated with at most one entity of entity set Y and vice versa.

Example: One student can register for numerous courses. However, all those courses have a single line back to that one student.



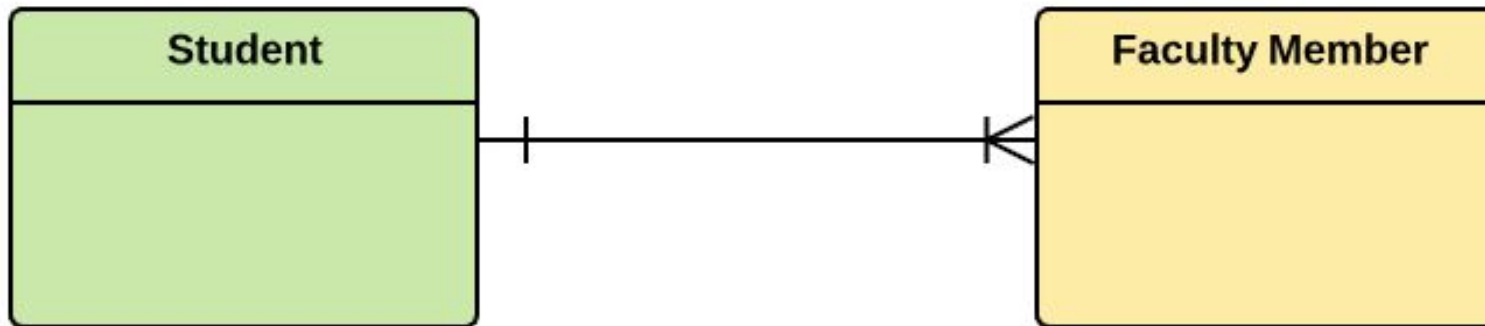
2. One-to-many & **Many to One**

- ▶ One entity from entity set X can be associated with multiple entities of entity set Y, but an entity from entity set Y can be associated with at least one entity.
- ▶ For example, one class is consisting of multiple students.
- ▶ **Many to One**
- ▶ More than one entity from entity set X can be associated with at most one entity of entity set Y. However, an entity from entity set Y may or may not be associated with more than one entity from entity set X.

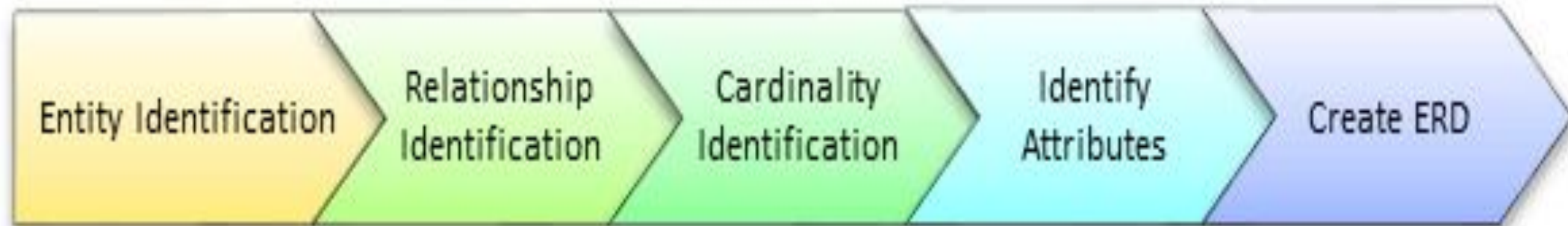


Many to Many:

- ▶ One entity from X can be associated with more than one entity from Y and vice versa.
- ▶ For example, Students as a group are associated with multiple faculty members, and faculty members can be associated with multiple students.



How to Create an Entity Relationship Diagram (ERD)

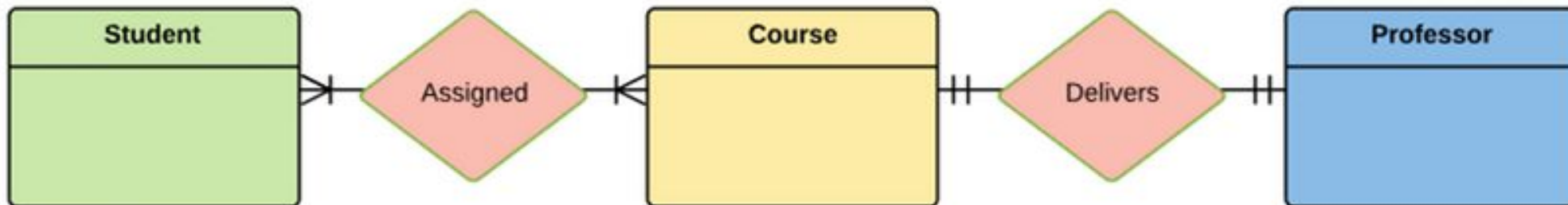


Let's study them with an Entity Relationship Diagram Example:

- ▶ **Step 1) Entity Identification** :- Student Course Professor
- ▶ **Step 2) Relationship Identification** : The student is **assigned** a course
 , Professor **delivers** a course

Step 3) Cardinality Identification

- ▶ A student can be assigned **multiple** courses
A Professor can deliver only **one** course

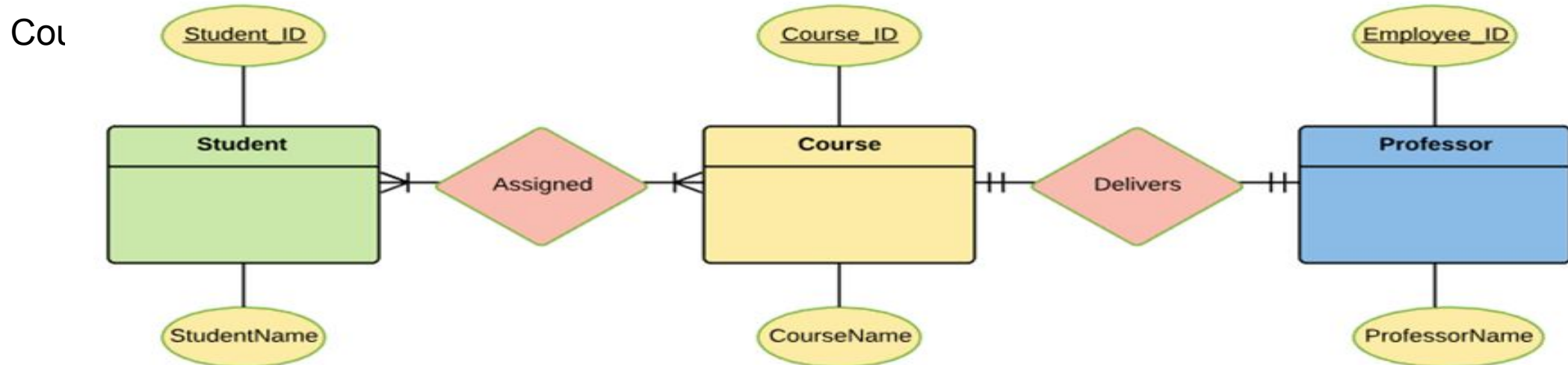


Step 4) Identify Attributes

- Once, you have a list of Attributes, you need to map them to the identified entities. Ensure an attribute is to be paired with exactly one entity.

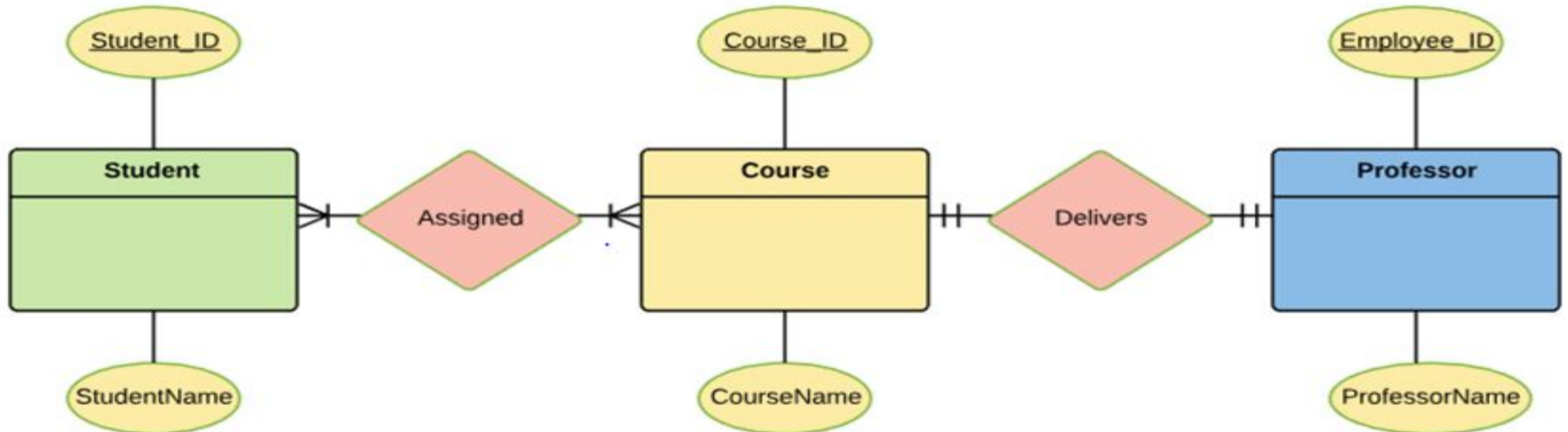
Once the mapping is done, identify the primary Keys. If a unique key is not readily available, create one.

Entity	Primary Key	Attribute
Student	Student_ID	StudentName
Professor	Employee ID	ProfessorName



Step 5) Create the ERD Diagram

- ▶ A more modern representation of Entity Relationship Diagram Example



Bank-details to construct ER-diagram

Bank have Customer.

Banks are identified by a name, code, address of main office.

Banks have branches.

Branches are identified by a branch_no., branch_name, address.

Customers are identified by name, cust-id, phone number, address.

Customer can have one or more accounts.

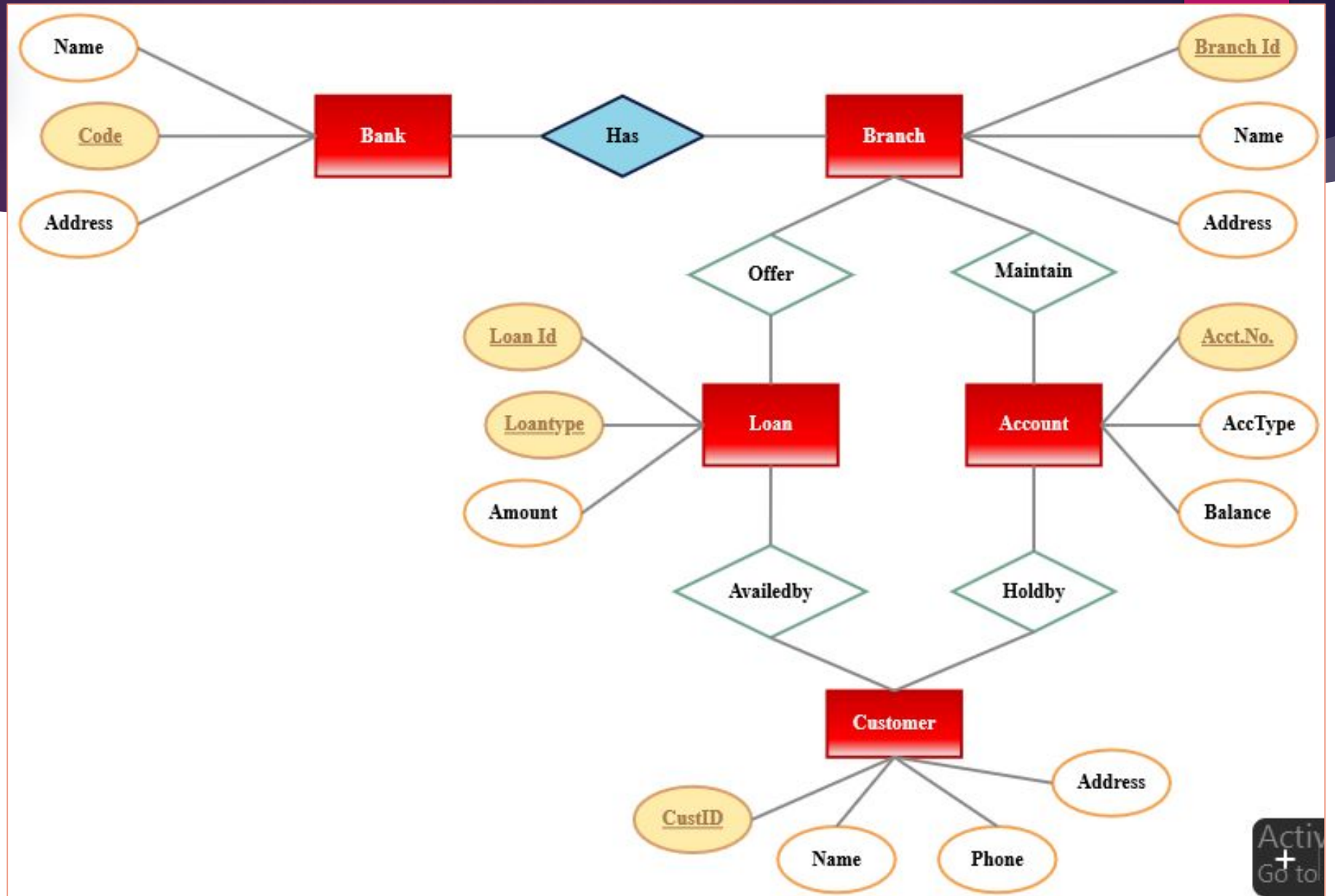
Accounts are identified by account_no., acc_type, balance.

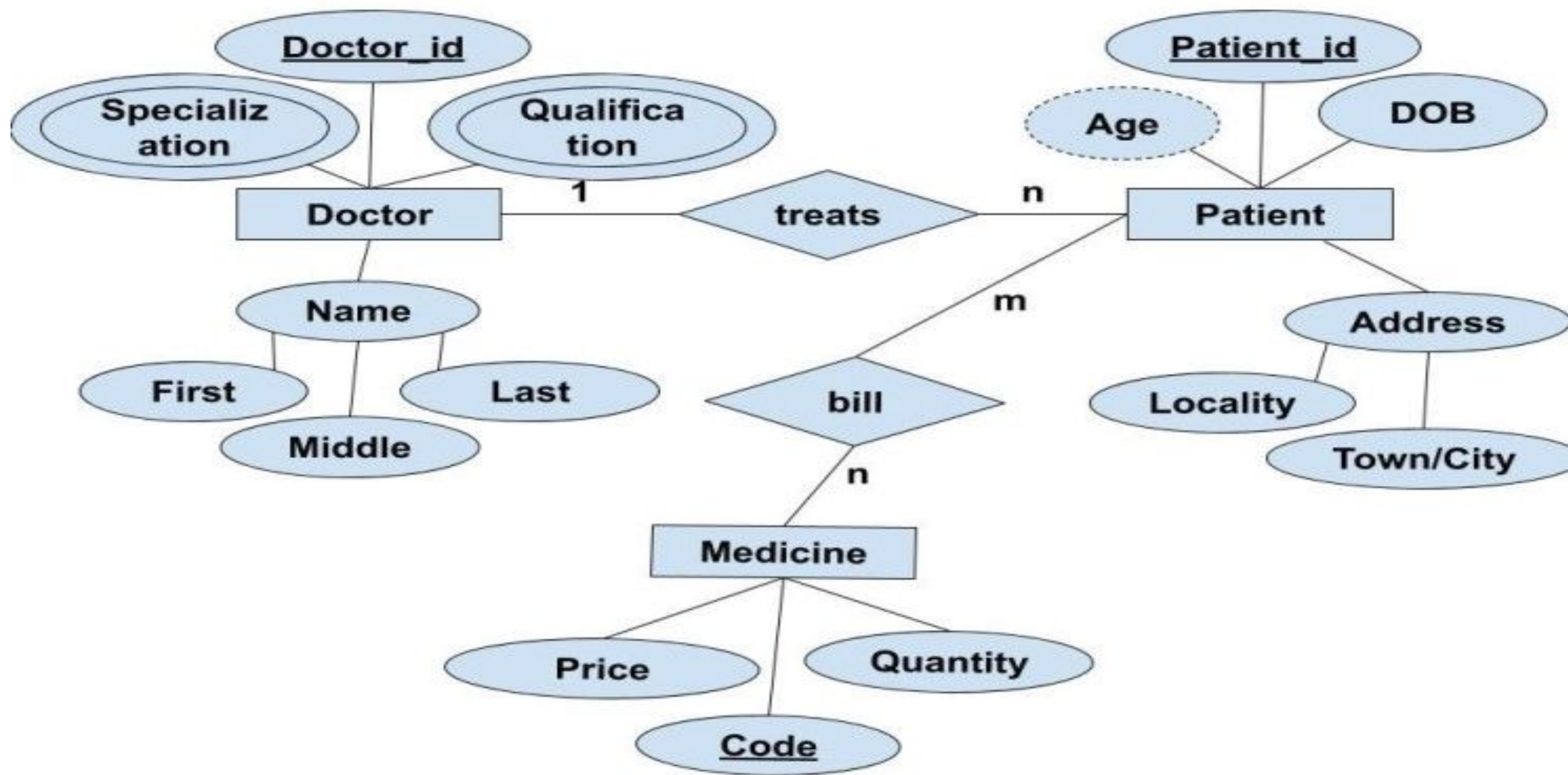
Customer can avail loans.

Loans are identified by loan_id, loan_type and amount.

Account and loans are related to bank's branch.

E-R diagram for Banking System





Hospital Management System

How to Draw an ER Diagram?

- ▶ **CustomerID:** A unique identifier for each customer registered on the platform.
- ▶ **Name:** The name of the customer.
- ▶ **Email:** The email address of the customer.
- ▶ **2. Product**
- ▶ This entity represents the products available for purchase on the platform.
- ▶ **Attributes**
- ▶ **ProductID:** A unique identifier for each product listed on the platform.
- ▶ **Name:** The name of the product.
- ▶ **Price:** The price of the product.

3. Order

This entity represents the orders placed by customers on the platform.

Attributes

OrderID: A unique identifier for each order placed on the platform.

OrderDate: The date on which the order was placed.

CustomerID (Foreign Key): Reference to the CustomerID in the Customer entity to identify who placed the order.

ProductID (Foreign Key): Reference to the ProductID in the Product entity to identify which product is ordered.

Quantity: The number of units of the product ordered.

Relationships

▶ 1. Places

- ▶ This is a relationship between the Customer and Order entities.
- ▶ Customers can place multiple orders, thus creating a one-to-many relationship from Customer to Order.

▶ 2. Contains

- ▶ This is a relationship between the Order and Product entities.
- ▶ An order can contain multiple products; similarly, a product can be contained in multiple orders, establishing a many-to-many relationship.
- ▶ This ER model gives a foundational view of how an e-commerce database might be structured, facilitating various operations like order tracking, inventory management, and customer relationship management. It helps understand how data is stored, retrieved, and managed in an e-commerce environment.

E-R diagram Customer- Order

